# Midterm Project- R Code and Outputs

### Load in neccessary packages

```
library(tidymodels)
library(caret)
library(corrplot)
library(glmnet)
library(kableExtra)
```

## Part I

Using this dataset, please help Researcher A build a prediction model for PFS, aiming to understand how demographic, tumor, and treatment characteristics influence progression-free survival in breast cancer patients.

To answer this question, your report should at least include the following:

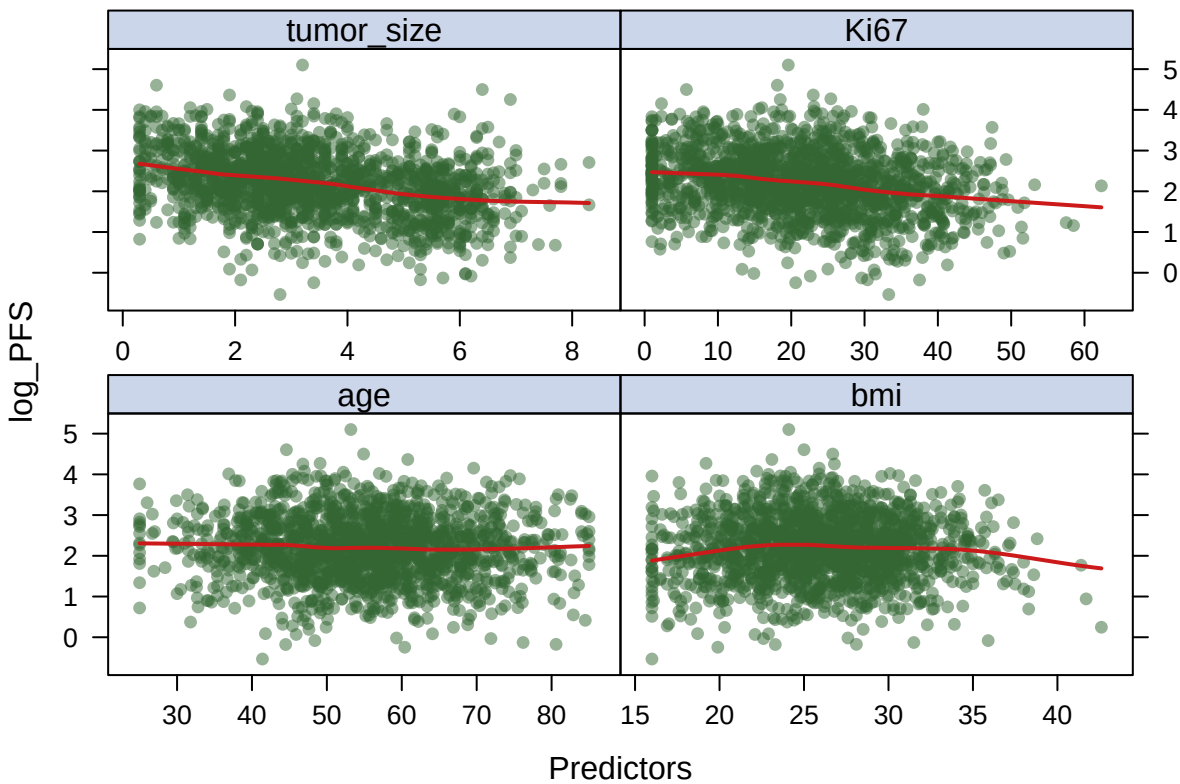
Model Training: Provide a detailed description of the model training procedure and how you obtained the model. Your description should include sufficient detail so that another statistician can reproduce the same model. Points will be deducted if the details provided are insufficient to reproduce your results.

### Import and partition data using 'tidymodels' package

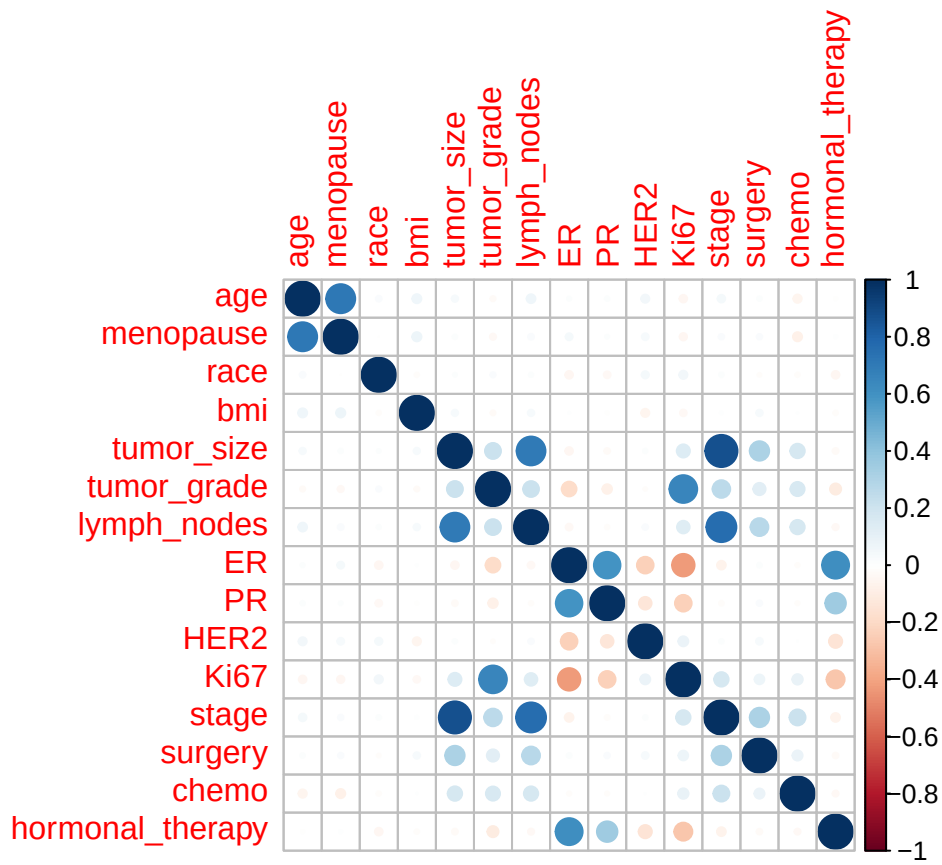
```
load("data/datA.RData")

set.seed(2222)

datA_split = initial_split(datA, prop = 0.8)

#extract the training and test data
training_datA = training(datA_split)
testing_datA = testing(datA_split)

#training data
x_trainA = model.matrix(log_PFS ~ . - id, training_datA)[, -1]
y_trainA = training_datA$log_PFS

#test data
x_testA = model.matrix(log_PFS ~ . - id, testing_datA)[, -1]
y_testA = testing_datA$log_PFS
```

### Exploratory Data Analysis

```
# Changing the plot display
theme1 <- trellis.par.get()
theme1$plot.symbol$col <- rgb(.2, .4, .2, .5)
theme1$plot.symbol$pch <- 16
theme1$plot.line$col <- rgb(.8, .1, .1, 1)
theme1$plot.line$lwd <- 2
theme1$strip.background$col <- rgb(.0, .2, .6, .2)
trellis.par.set(theme1)

# Feature plots to look at trends between predictors and the outcome (log_PFS)
featurePlot(x = training_datA[ ,c(2, 5, 6, 12)],
            y = training_datA[ ,17],
            plot = "scatter",
            span = .5,
            labels = c("Predictors","log_PFS"),
            type = c("p", "smooth"),
            layout = c(2, 2))
```




```
# Calculates the correlation matrix for all the predictors
corrplot(cor(x_trainA), method = "circle", type = "full")
```




This data set includes 7 binary predictors, 3 categorical predictors, 5 continuous predictors, and 1 continuous outcome in log scale.

Chose elastic net because continuous variables formed a cloud, meaning we could assume linearity? homoscedasticity?

Correlation plot revealed that certain variables were highly correlated. Chose to address this with elastic net because we can address colinearity for [] and [] reasons

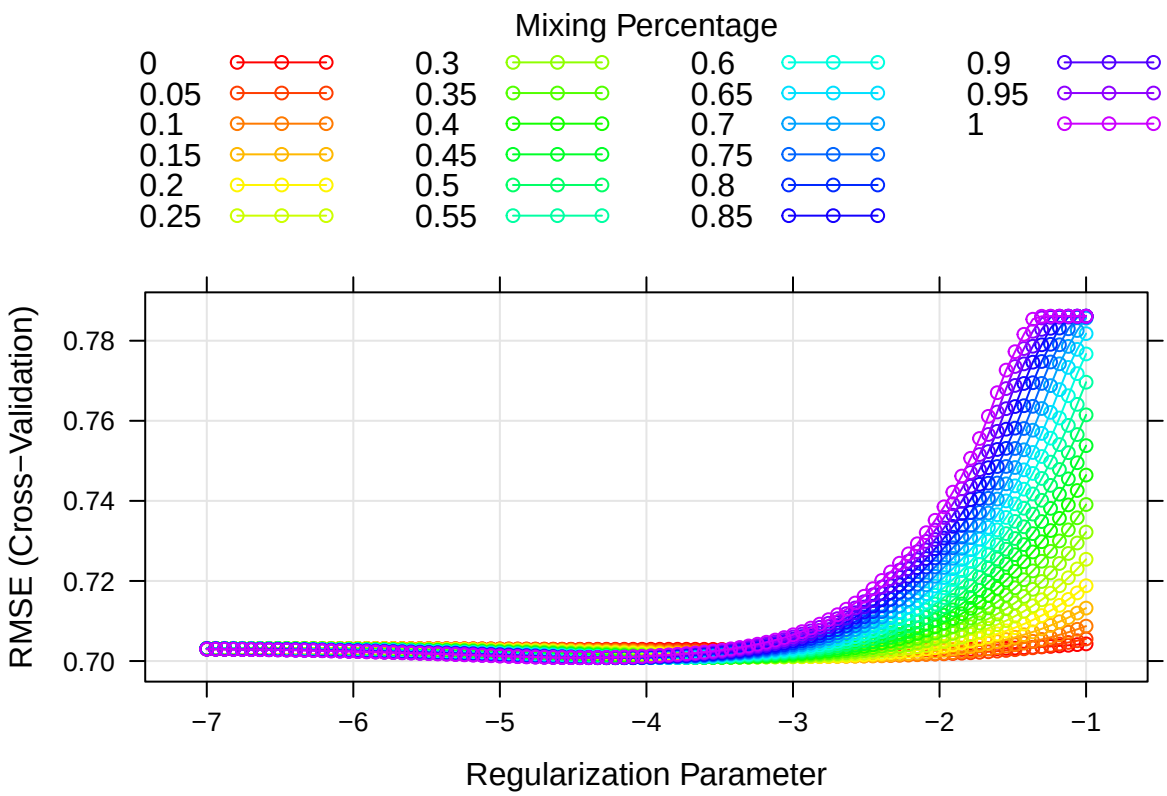
```
# Elastic Net

ctrl1 <- trainControl(method = "cv", number = 10) # set up cv
set.seed(2) # set seed for reproduction

# remove ID
training_datA <- training_datA |>  select(-"id")
testing_datA  <- testing_datA  |> select(-"id")

enet.fit <- train(log_PFS ~ .,
                  data = training_datA,
                  method = "glmnet",
                  tuneGrid = expand.grid(alpha = seq(0, 1, length = 21),
                                         lambda = exp(seq(-1, -7, length = 100))),
                  trControl = ctrl1)
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```r
enet.fit$bestTune
```

```
##      alpha     lambda
## 1944  0.95 0.01235197
```

```r
myCol <- rainbow(25)
myPar <- list(superpose.symbol = list(col = myCol),
superpose.line = list(col = myCol))

plot(enet.fit, par.settings = myPar, xTrans = log)
```




```r
# coefficients in the final model
coef(enet.fit$finalModel, enet.fit$bestTune$lambda)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##                   s=0.01235197
## (Intercept)        2.656050011
## age               -0.001046560
## menopause          .
## race               .
## bmi                .
## tumor_size        -0.057703954
## tumor_grade       -0.066041243
## lymph_nodes       -0.056993530
```

```
## ER                 0.220684085
## PR                 .
## HER2              -0.176250264
## Ki67              -0.003932773
## stage              .
## surgery            .
## chemo              .
## hormonal_therapy   0.137999051
```

Still to do: predict on testing_datA

Results: Report the model. Interpret the model and assess its performance.

```r
# Create a table with the model coefficients
coefs1 <- as.data.frame(as.matrix(coef(enet.fit$finalModel, enet.fit$bestTune$lambda)))
coefs1$Predictors <- rownames(coefs1)
colnames(coefs1)[1] <- "Coefficients"
coefs1 <- coefs1[, c("Predictors", "Coefficients")]
rownames(coefs1) <- NULL
coefs1 |>
  kable()
```

| Predictors | Coefficients |
|---|---|
| (Intercept) | 2.6560500 |
| age | -0.0010466 |
| menopause | 0.0000000 |
| race | 0.0000000 |
| bmi | 0.0000000 |
| tumor_size | -0.0577040 |
| tumor_grade | -0.0660412 |
| lymph_nodes | -0.0569935 |
| ER | 0.2206841 |
| PR | 0.0000000 |
| HER2 | -0.1762503 |
| Ki67 | -0.0039328 |
| stage | 0.0000000 |
| surgery | 0.0000000 |
| chemo | 0.0000000 |
| hormonal_therapy | 0.1379991 |

```r
# Create a table with the model coefficients exponentiated
coefs2 <- as.data.frame(as.matrix(coef(enet.fit$finalModel, enet.fit$bestTune$lambda)))
coefs2$Predictors <- rownames(coefs2)
colnames(coefs2)[1] <- "Exponentiated Coefficients"
coefs2 <- coefs2[, c("Predictors", "Exponentiated Coefficients")]
coefs2 <- coefs2 |>
  filter(Predictors %in% c("(Intercept)", "age", "tumor_size", "tumor_grade", "lymph_nodes",
                           "ER", "HER2", "Ki67", "hormonal_therapy"))
coefs2$`Exponentiated Coefficients`<- exp(coefs2$`Exponentiated Coefficients`)
rownames(coefs2) <- NULL
coefs2 |>
  kable()
```

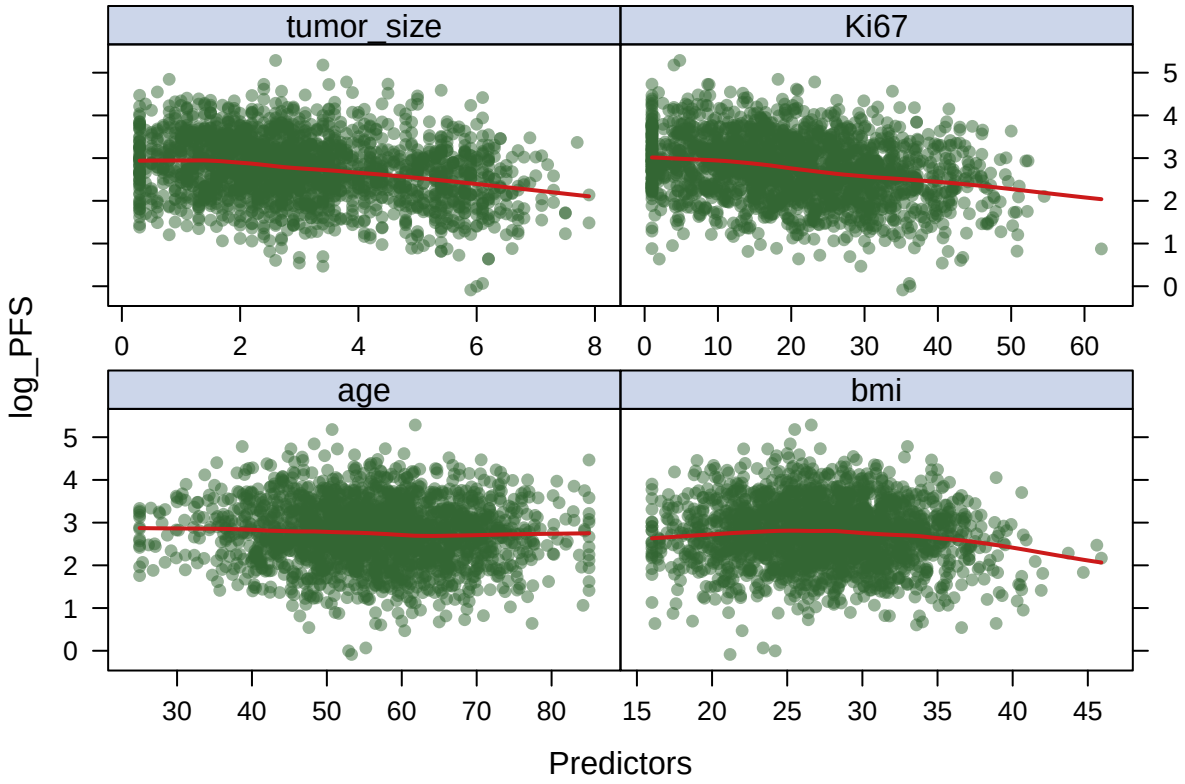
| Predictors | Exponentiated Coefficients |
|---|---|
| (Intercept) | 14.2399303 |
| age | 0.9989540 |
| tumor_size | 0.9439294 |
| tumor_grade | 0.9360923 |
| lymph_nodes | 0.9446002 |
| ER | 1.2469294 |
| HER2 | 0.8384081 |
| Ki67 | 0.9960750 |
| hormonal_therapy | 1.1479745 |

# Part II

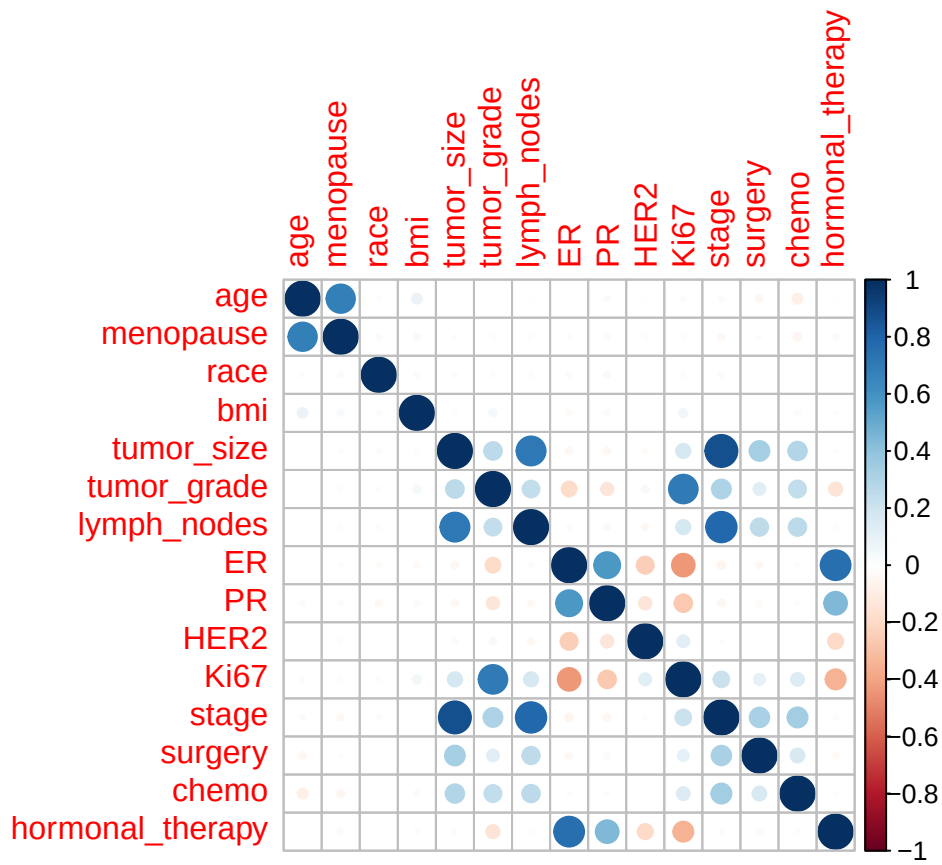
About a decade later, Researcher B conducted a separate study enrolling 2000 breast cancer patients diagnosed between 2000 and 2005, using the same study design and variable definitions. Like Researcher A's study, all patients were followed until disease progression or death. Researcher B's data are stored in datB.RData. Researcher A would like to evaluate whether the prediction model built on the original data generalizes well to Researcher B's dataset. Does the model generalize well? If yes, provide justification. If not, build a new prediction model for the 2000–2005 cohort and discuss what has changed in prediction.

## Import and explore the full dataset

```r
load("data/datB.RData")

# remove id for matrix operations
datB_clean_eda <- datB |> select(-"id")
x_B_all <- model.matrix(log_PFS ~ ., datB_clean_eda)[, -1]

# feature plots on full datB
featurePlot(x = datB[ , c(2, 5, 6, 12)],
            y = datB[ , 17],
            plot   = "scatter",
            span   = .5,
            labels = c("Predictors", "log_PFS"),
            type   = c("p", "smooth"),
            layout = c(2, 2))
```




```
# correlation plot on full datB
corrplot(cor(x_B_all), method = "circle", type = "full")
```




### Import and evaluate the model's performance on researcher B's dataset

```
# remove ID
datB <- datB |>  select(-"id")

# use all of datB to test generalizability of elastic net model
enet_pred_B <- predict(enet.fit, newdata = datB)

# compare performance (will update once model is tested on datA testing split)
# rmse_A <- RMSE(enet_pred_A,      testing_datA$log_PFS)
# r2_A   <- R2(enet_pred_A,        testing_datA$log_PFS)
rmse_B <- RMSE(enet_pred_B, datB$log_PFS)
r2_B   <- R2(enet_pred_B,   datB$log_PFS)

# cat("datA Test RMSE:", round(rmse_A, 4), "\n")
# cat("datA Test R²:  ", round(r2_A,   4), "\n\n")
cat("datB RMSE:     ", round(rmse_B, 4), "\n")
```

```
## datB RMSE:      0.8383
```

```
cat("datB R²:       ", round(r2_B,   4), "\n")
```

```
## datB R²:        0.1489
```

```r
# will add adjusted r^2
```

## TO DO ONLY IF NEED TO CREATE NEW MODEL

### Import and partition data using 'tidymodels' package

```r
set.seed(2222)

datB_split = initial_split(datB, prop = 0.8)

#extract the training and test data
training_datB = training(datB_split)
testing_datB = testing(datB_split)

#training data
x_trainB = model.matrix(log_PFS ~ ., training_datB)[, -1]
y_trainB = training_datB$log_PFS

#test data
x_testB  = model.matrix(log_PFS ~ ., testing_datB)[, -1]
y_testB = testing_datB$log_PFS
```

### Exploratory Data Analysis

```r
# Feature plots to look at trends between predictors and the outcome (log_PFS)
# featurePlot(x = training_datB[ ,2:16],
#             y = training_datB[ ,17],
#             plot = "scatter",
#             span = .5,
#             labels = c("Predictors","log_PFS"),
#             type = c("p", "smooth"),
#             layout = c(3, 2))
#
# # Calculates the correlation matrix for all the predictors
# corrplot(cor(x_trainB), method = "circle", type = "full")
```

Run GAM or MARs if model isn't super generalizable?